Estimation and Limit Theorems

50 minutes • 50 points • Instructor key

Grading note. Award method credit for correct reasoning after an earlier error; avoid repeated deductions for the same arithmetic mistake. Suggested working time is 45 minutes, leaving 5 minutes for checking.

Instructions. Answer all four problems and show your reasoning. No calculator is needed; exact fractions are acceptable. Any needed normal probabilities are supplied. You may quote Chebyshev's inequality, but must derive the law of large numbers in Problem 2. Use the spaces provided; clearly label any work continued on the back. Reserve 5 minutes to check.

1. Estimating a mean and variance

10 points · about 8 minutes

A random sample of six observations from a population with finite mean μ and variance σ^2 gives the data

$$4, \quad 6, \quad 6, \quad 8, \quad 8, \quad 10.$$

(a) Estimate μ using the sample mean $\bar{x}$. Show your calculation. (3 points)

Solution. The sum is 42, so $\bar{x} = 42/6 = 7$.

Credit: Sum of the observations: 1; division by 6: 1; value: 1.

(b) Calculate the sum of squared deviations from your sample mean, $\sum_{i=1}^{6}(x_i - \bar{x})^2$. (3 points)

Solution. The deviations are $-3, -1, -1, 1, 1, 3$, so

$$\sum_{i=1}^{6}(x_i - \bar{x})^2 = 9 + 1 + 1 + 1 + 1 + 9 = 22.$$

Alternatively, $\sum x_i^2 = 316$ and $316 - 6 \cdot 7^2 = 22$.

Credit: Correct deviations or equivalent computational identity: 1; squared terms: 1; sum: 1.

(c) Estimate σ^2 using the *unbiased sample variance* (divide the sum in part (b) by $n - 1$). Also report the mean squared deviation (divide by n). Label the two answers. (4 points)

Solution. The unbiased variance estimate is $s^2 = 22/(6-1) = 22/5$. The mean squared deviation is $22/6 = 11/3$.

Credit: Unbiased estimate: 2; mean squared deviation: 2. Give full method credit for correctly using an earlier computed sum. A proof of unbiasedness is not required.

2. Chebyshev's law of large numbers

14 points · about 12 minutes

Let $X_1, X_2, \ldots$ be independent and identically distributed, with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Put $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

(a) Find $\mathbb{E}[\bar{X}_n]$ and $\mathrm{Var}(\bar{X}_n)$. Show the calculation, and identify where independence is used. (4 points)

Solution. By linearity, $\mathbb{E}[\bar{X}_n] = n^{-1}\sum_i \mathbb{E}[X_i] = \mu$. Independence makes the variances add, so

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

Linearity of expectation itself does not require independence.

Credit: Mean with calculation: 1; $1/n^2$ scaling: 1; variance addition justified by independence: 1; final variance: 1.

(b) Apply Chebyshev's inequality to prove that, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty.$$

State what this says about the mode of convergence of $\bar{X}_n$. (6 points)

Solution. Chebyshev and part (a) give

$$0 \leq \mathbb{P}\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

Here σ^2 is finite and ε is fixed and positive. Thus $\bar{X}_n \to \mu$ in probability, which is the weak law of large numbers. This argument does not establish almost-sure convergence.

Credit: Apply Chebyshev to the sample mean: 2; substitute σ^2/n: 1; limit for every fixed positive ε: 2; identify convergence in probability: 1.

(c) Suppose $\sigma^2 = 4$. Find the smallest integer n for which *the Chebyshev bound* guarantees the following. (4 points)

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \tfrac{1}{2}\big) \leq 0.05.$$

Solution. The bound is $4/[n(1/2)^2] = 16/n$. Solve $16/n \leq 1/20$ to obtain $n \geq 320$; the requested integer is 320. This is sufficient by Chebyshev, not necessarily the smallest sample size for the actual distribution.

Credit: Bound $16/n$: 2; solve inequality: 1; integer 320: 1.

3. The central limit theorem

12 points · about 11 minutes

(a) State the usual central limit theorem for i.i.d. random variables with mean μ and variance σ^2. Give the assumption on σ^2, the standardized random variable, and its limiting distribution. A proof is not required. (4 points)

Solution. If $0 < \sigma^2 < \infty$, then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1).$$

The convergence is in distribution; the individual observations need not have a normal distribution.

Credit: Finite positive variance: 1; correct centering: 1; correct scaling: 1; standard normal limit in distribution: 1.

(b) Independent service times $X_1, \ldots, X_{100}$ have the same exponential distribution, with mean 2 minutes and variance 4 minutes squared. Use the CLT to approximate

$$\mathbb{P}(1.6 \le \bar{X}_{100} \le 2.4).$$

You may use $\Phi(2) \approx 0.9772$ and $\Phi(-2) \approx 0.0228$, where $\Phi(z) = \mathbb{P}(Z \le z)$ for $Z \sim N(0,1)$. Show the standard error and both standardized endpoints. (5 points)

Solution. The standard error is $\sigma/\sqrt{n} = 2/10 = 0.2$ minutes. The endpoints become $(1.6 - 2)/0.2 = -2$ and $(2.4 - 2)/0.2 = 2$. Hence

$$\mathbb{P}(1.6 \le \bar{X}_{100} \le 2.4) \approx \Phi(2) - \Phi(-2) \approx 0.9772 - 0.0228 = 0.9544.$$

Credit: Standard error: 1; each standardized endpoint: 1; normal probability difference: 1; approximate value: 1. Also accept 0.9545 from more precise normal probabilities.

(c) Chebyshev's inequality gives

$$\mathbb{P}(|\bar{X}_{100} - 2| \ge 0.4) \le \frac{4}{100(0.4)^2} = 0.25.$$

Translate this into a lower bound for the interval probability in part (b). Explain why that bound and your CLT answer are compatible, and identify which is a guaranteed bound and which is an approximation. (3 points)

Solution. Chebyshev guarantees probability at least $1 - 0.25 = 0.75$ in the open interval $(1.6, 2.4)$, hence also in the closed interval in part (b). The CLT suggests about 0.9544, which is above this lower bound. Chebyshev is a valid finite-sample bound; the CLT supplies an asymptotic approximation, not an exact finite-sample probability or error guarantee.

Credit: Lower bound 0.75: 1; explain compatibility: 1; correctly distinguish guarantee from approximation: 1.

4. A heavy-tailed distribution

14 points · about 14 minutes

Let $X_1, X_2, \dots$ be i.i.d. with density

$$f(x) = \begin{cases} \dfrac{1}{2x^{3/2}}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

This is a Pareto distribution. Write $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $M_n = \max\{X_1, \dots, X_n\}$.

(a) Verify that f integrates to 1, and calculate $\mathbb{P}(X_1 > t)$ for $t \geq 1$. (4 points)

Solution. The density is nonnegative and

$$\int_1^\infty \frac{dx}{2x^{3/2}} = [-x^{-1/2}]_1^\infty = 1, \qquad \mathbb{P}(X_1 > t) = \int_t^\infty \frac{dx}{2x^{3/2}} = \frac{1}{\sqrt{t}}.$$

Credit: Normalization integral and value: 2; tail integral and value: 2.

(b) Evaluate the improper integral for $\mathbb{E}[X_1]$. Explain why the law proved in Problem 2 does not apply. (3 points)

Solution. Since $X_1 \geq 0$,

$$\mathbb{E}[X_1] = \int_1^\infty x\, f(x)\, dx = \frac{1}{2}\int_1^\infty x^{-1/2}\, dx = \lim_{R\to\infty}(\sqrt{R} - 1) = +\infty.$$

There is no finite population mean; the finite-moment hypotheses of the Chebyshev law are not satisfied.

Credit: Expectation integral: 1; divergence: 1; identify the missing finite mean: 1.

(c) Fix any $A > 0$ and take n large enough that $nA \geq 1$. Use independence and part (a) to find $\mathbb{P}(M_n \leq nA)$. (3 points)

Solution. All n observations must be at most nA, so

$$\mathbb{P}(M_n \leq nA) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq nA) = \left(1 - \frac{1}{\sqrt{nA}}\right)^n.$$

Credit: Intersection of all n events: 1; product justified by independence: 1; final expression: 1.

(d) Use part (c) to prove $\mathbb{P}(\bar{X}_n \leq A) \to 0$ for every fixed $A > 0$. What does this say about convergence of $\bar{X}_n$ to a finite constant? (4 points)

Hint: All observations are nonnegative, so $\bar{X}_n \geq M_n/n$. You may use $1 - u \leq e^{-u}$ for $0 \leq u \leq 1$.

Solution. The event $\{\bar{X}_n \leq A\}$ implies $M_n \leq nA$. Thus

$$0 \leq \mathbb{P}(\bar{X}_n \leq A) \leq \left(1 - \frac{1}{\sqrt{nA}}\right)^n \leq e^{-\sqrt{n/A}} \longrightarrow 0.$$

Therefore $\mathbb{P}(\bar{X}_n > A) \to 1$ for every fixed $A > 0$: the sample average tends to $+\infty$ in probability, and cannot converge in probability to any finite constant. Indeed, convergence to a finite c would give $\mathbb{P}(\bar{X}_n \leq A) \to 1$ for any $A > \max\{c, 0\}$.

Credit: Correct event inclusion and probability bound: 1; exponential bound with correct exponent: 1; limit: 1; no finite constant limit: 1.

Conceptual note. Infinite variance alone would not make the i.i.d. LLN fail: a finite absolute first moment is sufficient. Here the infinite mean is accompanied by an explicit proof of divergence. Part (b) alone establishes only that the theorem from Problem 2 is unavailable.